



Research Paper Presentation Template

A reusable ISISLab/UNISA Beamer theme for clear scientific storytelling

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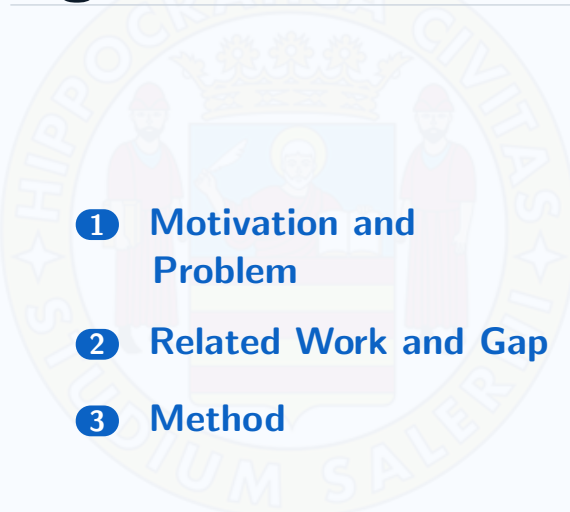
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Presenter Name · July 19, 2026

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Agenda

- 
- 1 Motivation and Problem
 - 2 Related Work and Gap
 - 3 Method
 - 4 Experimental Design
 - 5 Results
 - 6 Discussion and Conclusion





SECTION 1

Motivation and Problem

Define the research question and why it matters



Problem State the gap, audience, and practical consequence.

One clear research story

Narrative flow

- 1 Here is the problem
- 2 It matters to a clear audience
- 3 There is a gap in current work

Source guidance

A strong research paper centers on one reusable idea and makes the claim explicit. This demo follows the Microsoft Research writing structure: problem, idea, evidence, comparison, conclusion.^a

^aInspired by Simon Peyton Jones, *How to write a great research paper*, Microsoft Research, 2016.



Problem and contributions

Problem

Existing systems are accurate in controlled settings, but performance degrades when the data distribution changes. This creates unreliable outcomes in real deployments.

Research question

Can we design a method that remains accurate, efficient, and interpretable under realistic distribution shifts?

Contributions

- 1 A new robust learning formulation
- 2 A reproducible experimental protocol
- 3 Evidence across three benchmark scenarios





SECTION 02

Related Work and Gap

Position the contribution without turning the talk into a bibliography



Positioning

Compare families of approaches around the unresolved gap.

Where the paper fits

12

core papers

3

main families

Comparison logic

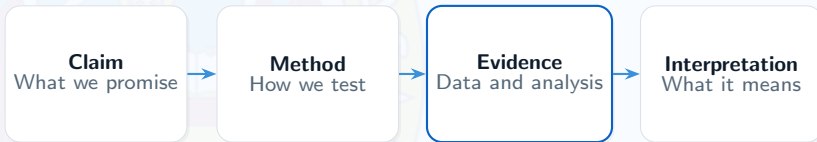
Avoid listing related work too early. First explain the problem and the idea; then compare alternatives around the gap the paper addresses.

Gap statement

Prior work optimizes either accuracy or robustness, but does not jointly report reliability, computational cost, and interpretability.



Claims must map to evidence



Evidence checklist

Every claim introduced in the first part of the presentation should be revisited with a figure, table, proof sketch, or experiment in the results section.





SECTION 03

Method

Move from intuition to a reproducible algorithm



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Core idea

Explain inputs, transformations, assumptions, and outputs.

Method pipeline



Design principle

Separate the core idea from implementation details. Show the intuition first, then the formal or algorithmic specification.

Method slide rule

One pipeline slide should explain how data becomes evidence. Later slides can zoom into each module.



Algorithmic components

Step 1

Represent each instance with stable and shift-sensitive features.

Step 2

Optimize a robust objective that penalizes unstable shortcuts.

Step 3

Explain decisions with component-level attribution.

Take-home message

The method is designed to preserve performance when the test environment differs from the training environment.





SECTION 04

Experimental Design



Make the evidence credible before showing the results

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Protocol

Specify data, baselines, metrics, controls, and reproducibility.

Experimental design

Evaluation matrix

- 1 Three datasets
- 2 Two distribution shifts
- 3 Accuracy and robustness metrics
- 4 Runtime and calibration

Protocol

Use the same train, validation, and test partitions for every baseline. Report average and variance across repeated runs.

3

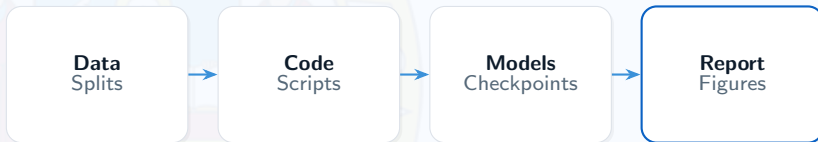
datasets

5

baselines



Reproducibility package



Artifact policy

Make the presentation point to the same artifacts as the paper: repository, data card, environment file, and evaluation scripts.





SECTION 05

Results

Lead with the main finding, then test why it holds



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Evidence Main result, ablation, uncertainty, and failure cases.

Main result: robustness improves



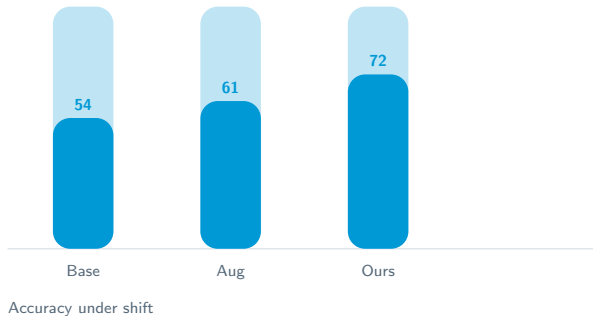
+18%

relative robustness gain

42 ms

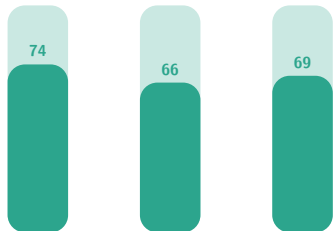
median latency

Demo bar chart



Ablation and training dynamics

Ablation



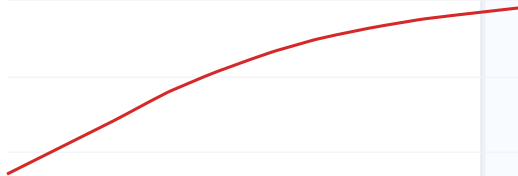
Robust accuracy

Full

NoReg

NoExpl

Trend



Validation score over epochs





SECTION 06

Discussion and Conclusion

Interpret the evidence and close the research story



Takeaway Limitations, implications, future work, and final message.

Limitations and threats to validity

Limitations

- 1 Only medium-scale datasets
- 2 Synthetic shift may not capture all deployment cases
- 3 Interpretability metric is task dependent

Mitigations

- 1 Release scripts for replication
- 2 Include diverse baselines
- 3 Report failure cases rather than hiding them



Conclusion

Answer to the research question

The method improves robustness under distribution shift while keeping runtime practical and explanations inspectable.

Future work

Scale to larger datasets, validate with field data, and add human-centered evaluation of explanations.

Final message

A great research presentation makes one idea memorable, states refutable contributions, and connects every result back to evidence.

Problem

Idea

Evidence

Impact



References

Cited template source

Simon Peyton Jones. *How to write a great research paper*. Microsoft Research, 2016.
microsoft.com/en-us/research/wp-content/uploads/2016/07/How-to-write-a-great-research-paper.pdf



Questions · Discussion · Next steps

